IGCSE 0607 Practice Questions — E4.4 — Combined Questions & Markscheme

Equations of linear graphs

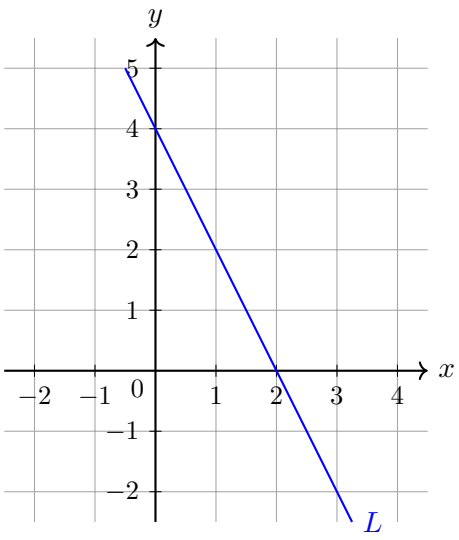
Total Questions: 6 — **Total Marks:** 40 Marks

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  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

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1 The diagram shows a straight line L drawn on a coordinate grid.



(a) Find the equation of line L.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(b) Write the equation of line L in the form $ax + by = c$, where a, b and c are integers.

..... **[1]**

2 **(a)** Find the gradient and the y-intercept of the line $3x + 5y = 15$.

Gradient = ... , y-intercept = ... **[3]**

(b) Find the equation of the line parallel to $3x + 5y = 15$ that passes through the point $(0, -4)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

3 The line L passes through the points $A(2a, a+4)$ and $B(a-1, 3a+2)$.
The gradient of line L is -4.

(a) Find the value of a.

$a =$ ..... **[3]**

(b) Find the equation of line L.
Give your answer in the form $px + qy = r$, where p, q and r are integers and $p > 0$.

..... **[3]**

(c) Line L intersects the line $y = 3x - 2$ at the point M.
Find the coordinates of M.

(..... , ...) **[3]**

4 A is the point $(-3, 1)$ and B is the point $(5, 5)$.

(a) Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

(b) The perpendicular bisector of AB intersects the line $x - 2y = 5$ at the point P.
Find the coordinates of P.

(..... , ...) **[3]**

(c) Find the exact length of AP.

..... **[3]**

5 Point A has coordinates $(-4, 3)$ and point B has coordinates $(6, -2)$.

(a) Find the equation of the line passing through A and B.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

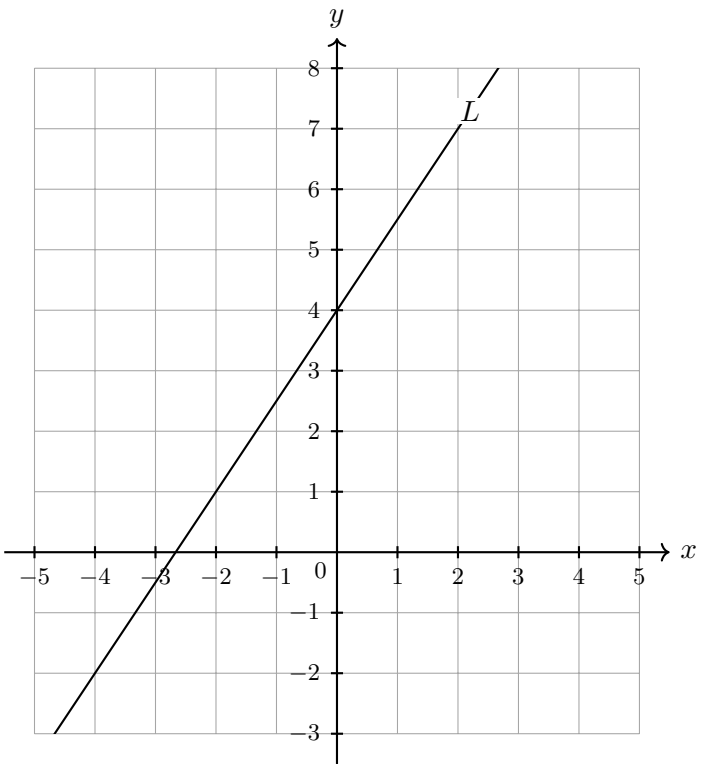
(b) The line $3x - 5y = 15$ intersects the x-axis at point C and the y-axis at point D.
Find the coordinates of C and D.

$C =$ ..... , $D =$ ..... **[2]**

(c) Write down the equation of the line that is parallel to the y-axis and passes through point A.

..... **[1]**

6 The diagram shows a straight line L.



(a) Find the equation of line L.
Give your answer in the form $ax + by = c$, where a, b, and c are integers.

..... **[3]**

(b) Line M is perpendicular to line L and passes through the point $(9, -1)$.
Find the equation of line M in the form $y = mx + c$.

$y =$ ..... **[3]**

1 **(a)** Gradient $= -2$ **[B1]**

y-intercept $= 4$ **[B1]**

$-2x + 4$ **[B1]**

(b) $2x + y = 4$ oe **[B1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** Rearranges equation to $y = mx + c$ form: $5y = -3x + 15$ or $y = -\frac{3}{5}x + 3$ **[M1]**

Gradient $= -\frac{3}{5}$ **[A1]**

y-intercept $= 3$ **[A1]**

(b) Identifies parallel gradient as $-\frac{3}{5}$ **[M1]**

$-\frac{3}{5}x - 4$ **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** $\frac{(3a+2)-(a+4)}{(a-1)-2a} = -4$ **[M1]**

$\frac{2a-2}{-a-1} = -4$ **[M1]**

$2a - 2 = 4a + 4$

-6 = 2a

a = -3[$A1$]

(b) Identifies coordinates of A as $(-6, 1)$ or B as $(-4, -7)$ **[M1]**

$y - 1 = -4(x + 6)$ or $y = -4x - 23$ **[M1]**

$4x + y = -23$ **[A1]**

(c) $4x + (3x - 2) = -23$ or $-4x - 23 = 3x - 2$ oe **[M1]**

$7x = -21$

x = -3[$A1$]

$y = -11$ **[A1]**

[Back to Question] *[Total Marks: 9]*

4 **(a)** Calculates the midpoint of $AB = \left(\frac{-3+5}{2}, \frac{1+5}{2}\right) = (1, 3)$ **[B1]**

Calculates the gradient of $AB = \frac{5-1}{5-(-3)} = \frac{4}{8} = \frac{1}{2}$ **[B1]**

Finds the perpendicular gradient $= -2$ **[M1]**

Uses perpendicular gradient -2 and midpoint $(1, 3)$ to find equation, e.g., $3 = -2(1) + c$ **[M1]**

$-2x + 5$ **[A1]**

(b) Sets up simultaneous equations, e.g., substituting $y = -2x + 5$ into $x - 2y = 5$ **[M1]**

$x - 2(-2x + 5) = 5 \implies 5x - 10 = 5 \implies x = 3$ **[M1]**

$(3, -1)$ **[A1]**

(c) Uses distance formula for $A(-3, 1)$ and $P(3, -1)$ **[M1]**

$\sqrt{(3 - (-3))^2 + (-1 - 1)^2}$ **[M1]**

$\sqrt{40}$ or $2\sqrt{10}$ **[A1]**

[Back to Question] *[Total Marks: 10]*

5 (a) Gradient $= \dfrac{-2 - 3}{6 - (-4)}$ or $\dfrac{-5}{10} = -\dfrac{1}{2}$ **[M1]**

Substitution of $(-4, 3)$ or $(6, -2)$ into $y = -\dfrac{1}{2}x + c$, e.g. $3 = -\dfrac{1}{2}(-4) + c$ **[M1]**

$-\dfrac{1}{2}x + 1$ **[A1]**

(b) $C = (5, 0)$ **[B1]**

$D = (0, -3)$ **[B1]**

(c) $x = -4$ **[B1]**

[Back to Question] *[Total Marks: 6]*

6 (a) Identifies two correct points on the line, e.g., $(0, 4)$ and $(2, 7)$ **[M1]**

Calculates the gradient $m = \dfrac{7 - 4}{2 - 0} = \dfrac{3}{2}$ **[M1]**

States correct equation in required integer form, e.g., $3x - 2y = -8$ or $2y - 3x = 8$ **[A1]**

(b) Uses $m_1 \times m_2 = -1$ to find gradient of M: $-\dfrac{2}{3}$ **[M1]**

Substitutes $(9, -1)$ into $y = -\dfrac{2}{3}x + c$ to find c: $-1 = -\dfrac{2}{3}(9) + c$ **[M1]**

$-\dfrac{2}{3}x + 5$ **[A1]**

[Back to Question] *[Total Marks: 6]*